📄 **Tech Stack Document**

🧠 **Project: HealMate (Medication Non-Adherence System)**

🔹 1. Overview

This document outlines the complete technology stack used to build the AI-powered medication adherence system, including frontend, backend, AI modules, and integrations.

🖥️ 2. Frontend (User Interface)

Technologies:

- React.js *(Web)* / Flutter *(Mobile - optional)*
- Tailwind CSS / Material UI

Features Implemented:

- Patient dashboard (medication schedule, reminders)
- Doctor dashboard (adherence analytics, risk levels)
- Real-time notifications UI
- Graphs & charts (Chart.js / Recharts)

⚙️ 3. Backend (Server-Side)

Technologies:

- Node.js with Express.js
- Firebase (alternative: backend + auth + hosting)

Responsibilities:

- API handling (REST APIs)
- User authentication & management
- Medication scheduling logic
- Data processing & storage

🧠 4. AI / Machine Learning Module

Technologies:

- Python
- scikit-learn / TensorFlow

Core Functions:

- Predict medication non-adherence
- Risk scoring (Low / Medium / High)
- Behavior pattern analysis

Input Data:

- Past medication logs
- Time-based adherence patterns

Output:

- Probability of missed dose
- Personalized alerts

🗄️ 5. Database

Technologies:

- MongoDB *(preferred)*

Stored Data:

- User profiles
- Medication schedules
- Adherence logs
- AI prediction results

📸 6. Computer Vision (Optional WOW Feature)

Technologies:

- OpenCV
- TensorFlow Lite / CNN model

Functionality:

- Pill detection via camera
- Verify correct medicine intake

🔔 7. Notifications & Alerts

Technologies:

- Firebase Cloud Messaging (FCM)

- Twilio API *(for SMS alerts)*

Features:

- Smart reminders
- Escalation alerts (caregiver notification)

🔐 8. Authentication & Security

Technologies:

- Firebase Auth / JWT Authentication

Features:

- Secure login/signup
- Protected health data
- Role-based access (Patient / Doctor)

☁️ 9. Deployment

Technologies:

- Frontend: Vercel / Netlify
- Backend: Render / Railway / Firebase
- Database: MongoDB Atlas

🔗 10. System Integration Flow

1. User inputs medication schedule
2. Backend stores data in database
3. AI model analyzes adherence patterns
4. Prediction generated
5. Notifications triggered
6. Dashboard updated in real-time

🏗️ 11. Architecture Summary

- **Frontend:** User interaction & visualization
- **Backend:** Logic + API + data handling
- **AI Layer:** Prediction & intelligence

- **Database:** Persistent storage
- **APIs:** Communication between components

🚀 12. Key Tech Highlights (For Judges)

- 🤖 AI-based prediction system
- 📊 Real-time analytics dashboard
- 📷 Computer vision integration (optional)
- 🔔 Intelligent alert system
- ☁️ Scalable cloud deployment